

Petri nets: Live and safe implies strongly connected

Heorhii Lopattn

June 7, 2025

Solution

In the network $N = \langle P, T, F \rangle$ Given that N is live, safe and connected, we will show that G - a graph constructed from N is strongly connected.

Let $v \in V(G)$, and N bounded with $m \in \mathbf{N}$. We need to show that the set S of vertices that reach v is equal to $V(G)$. Assume for the sake of contradiction it's not.

Since N is connected, there are edges from S to $U = V(G) \setminus S$. Take $u \in U$ such that there is an edge from $s \in S$ to u .

Either $u \in P, s \in T$ or $u \in T, s \in P$. Using liveness of N take an initial marking M and a sequence of transitions σ fireable from M such that u or s (depending on which is a transition) appear $m + 1$ times. Since there are no edges from U to S , all transitions in U can never activate transitions in S , therefore σ will still be a fireable sequence if we remove all transitions from S . Call this new sequence $\tilde{\sigma}$

Observe that $M[\tilde{\sigma}]$ has $m + 1$ tokens in u or s (depending on which is a place).

If u was a place, then we have fired s $m + 1$ times, so in total $m + 1$ tokens appeared in u . None of them could have left u , since all transitions are in U .

If s was a place, then there must be $m + 1$ tokens in s after firing $\tilde{\sigma}$ because we were supposed to fire u $m + 1$ times and we didn't. This contradicts boundness of N on M , so G is strongly connected.